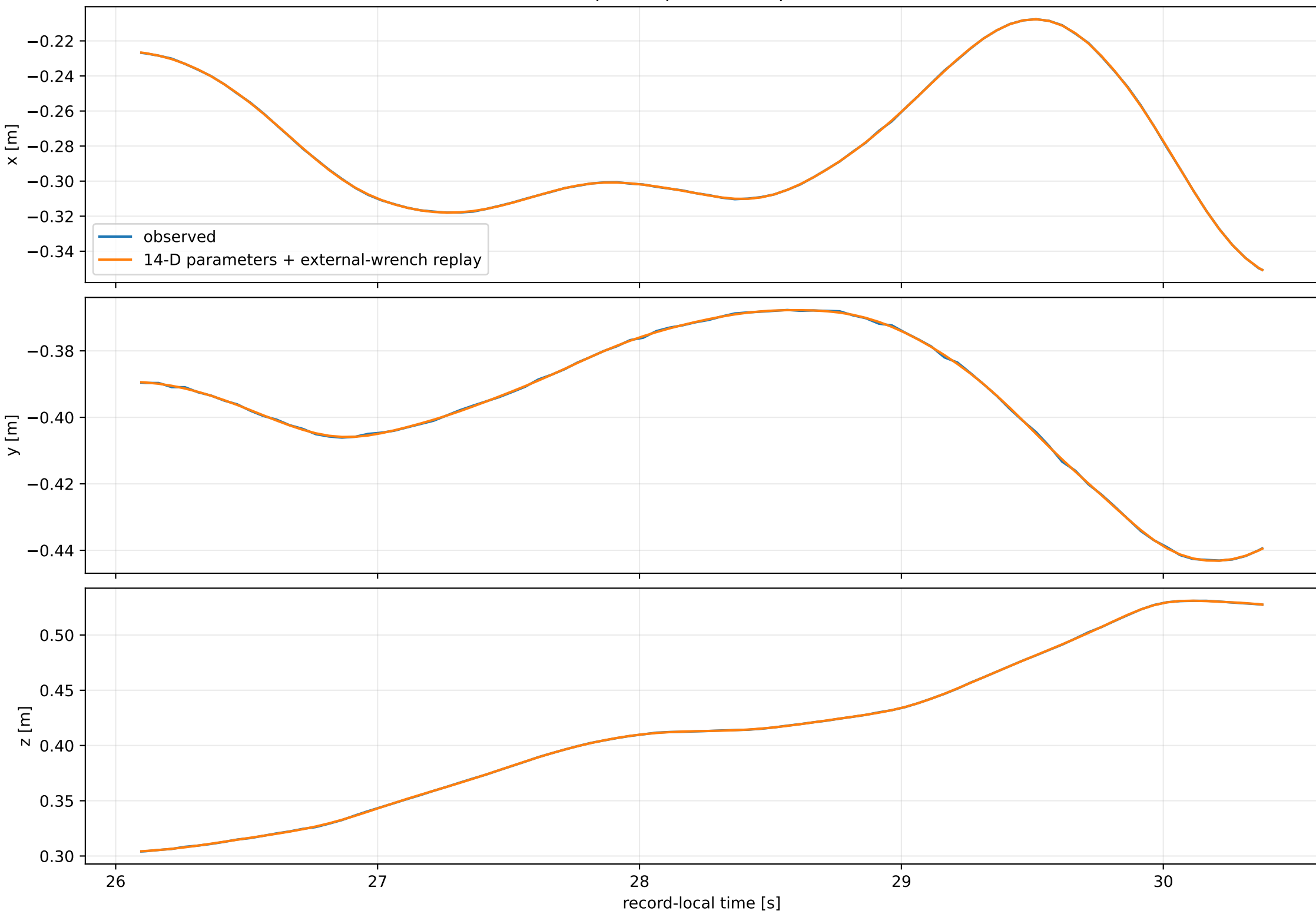
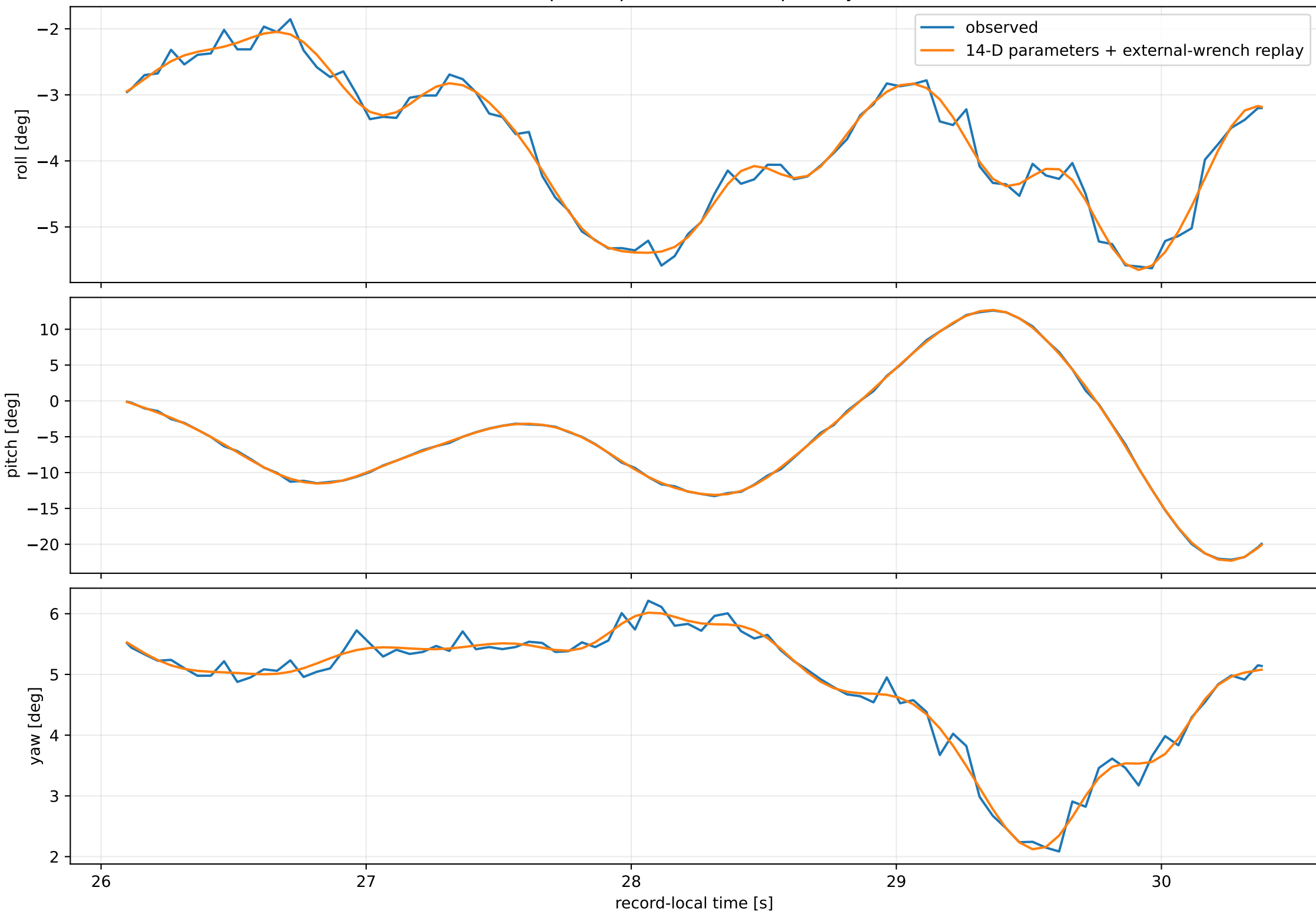


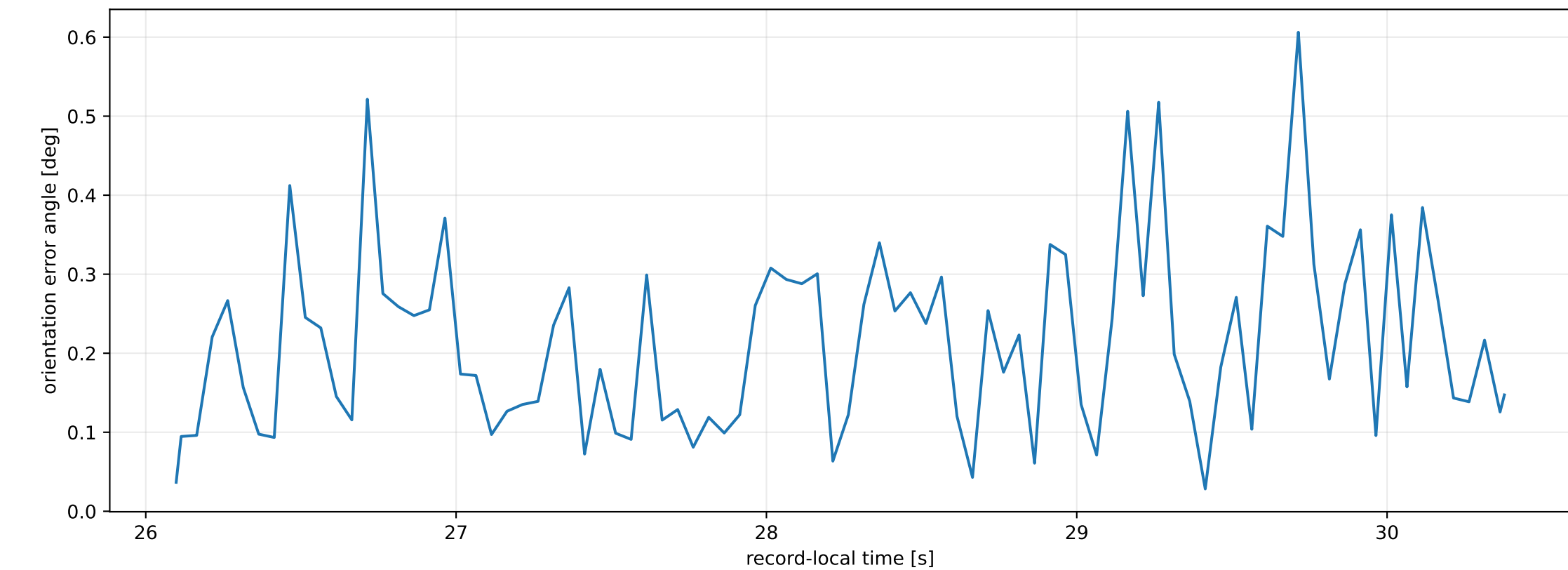
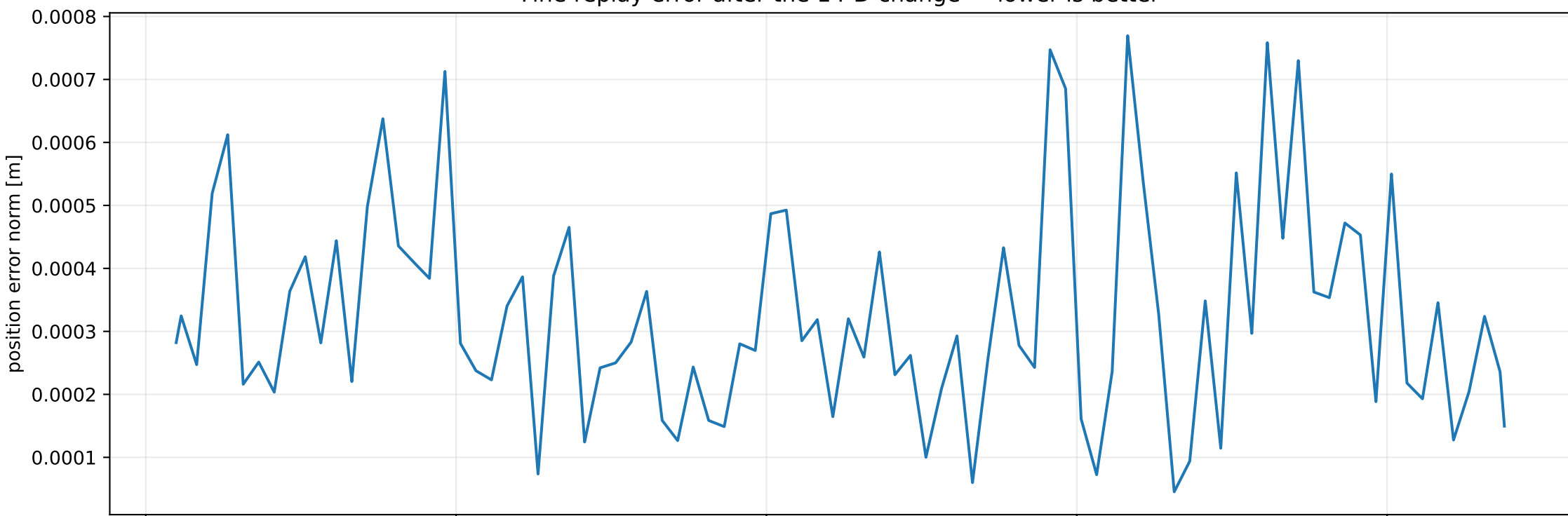
Fine pose reproduction: position



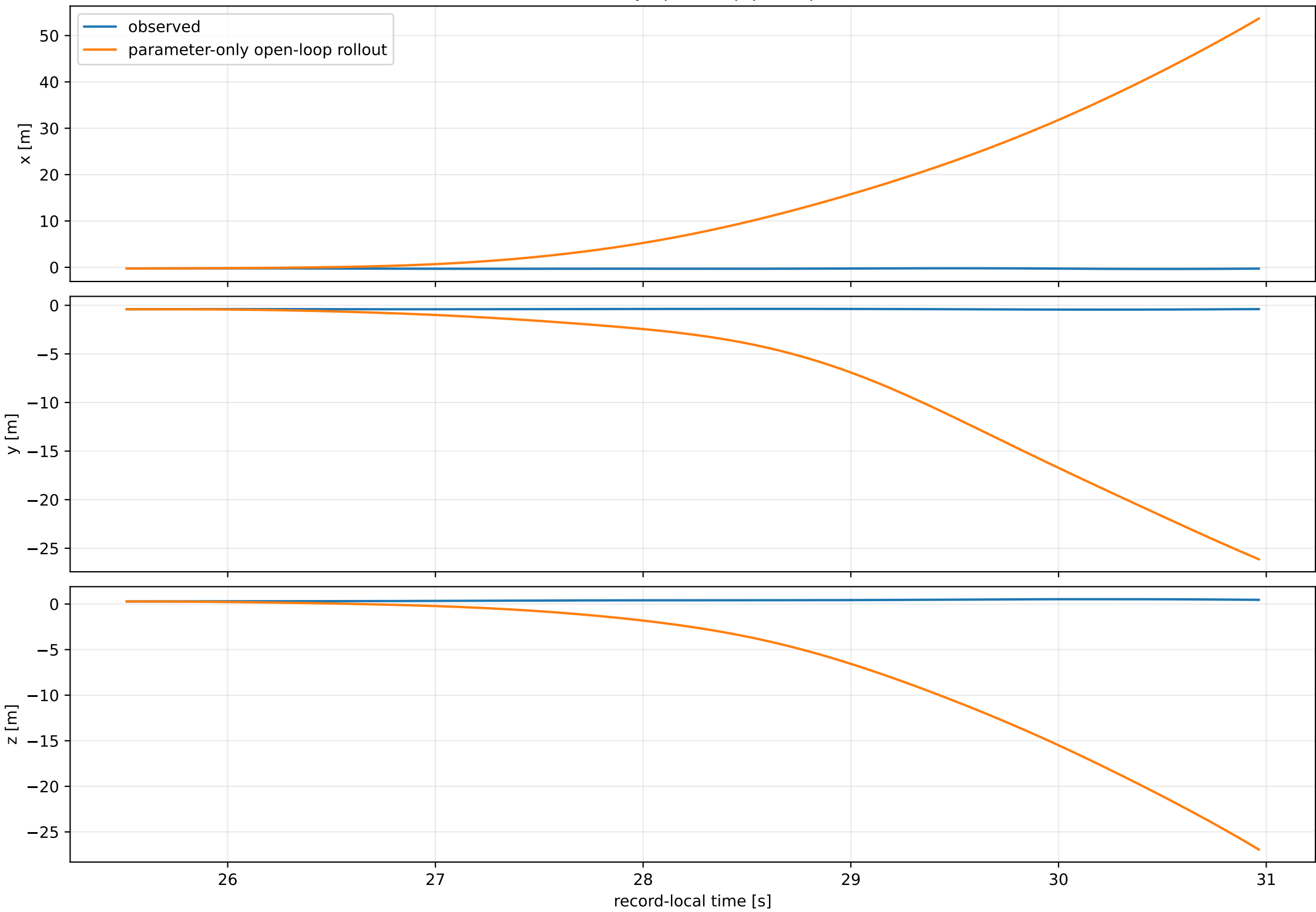
Fine pose reproduction: roll / pitch / yaw



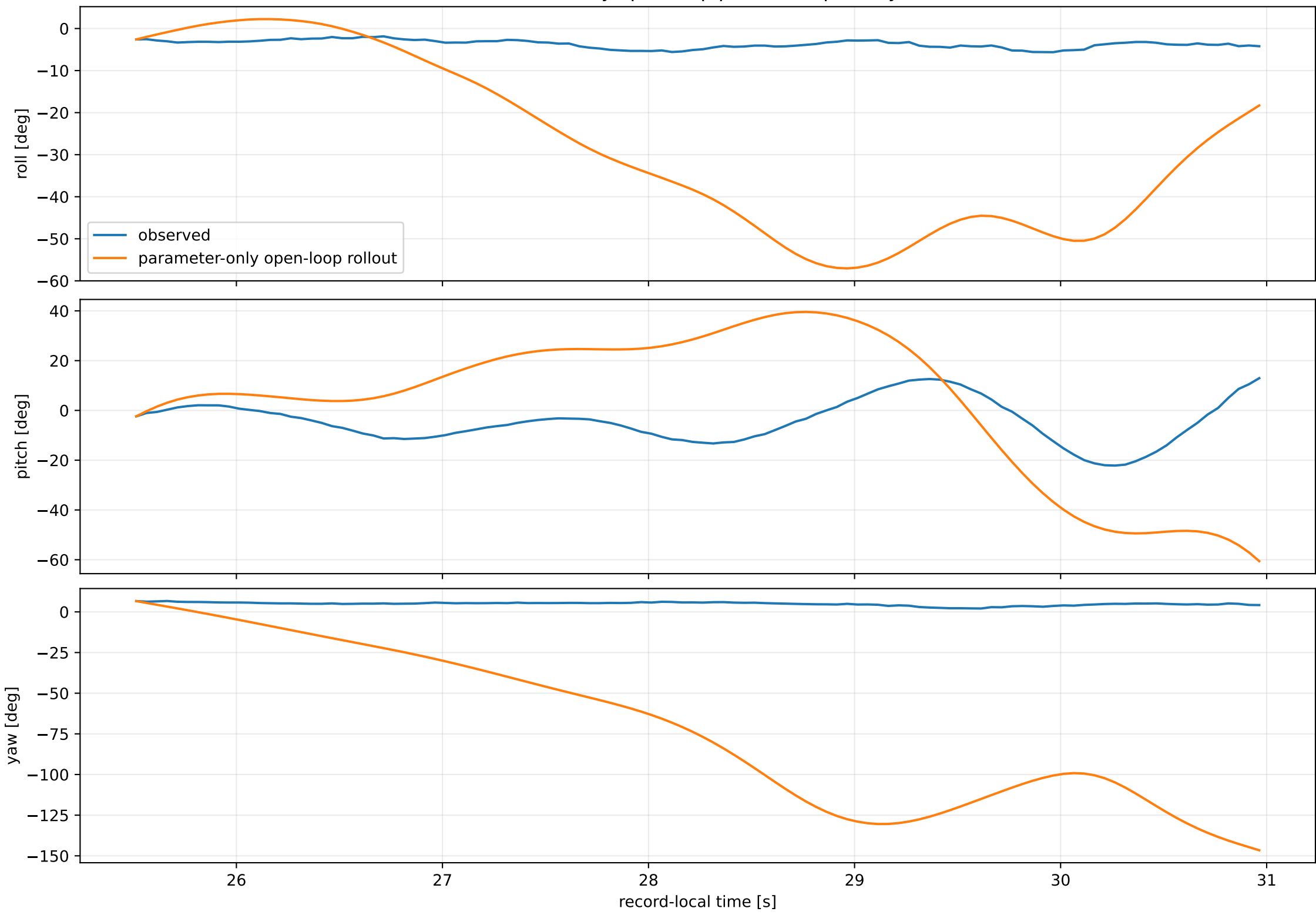
Fine replay error after the 14-D change — lower is better



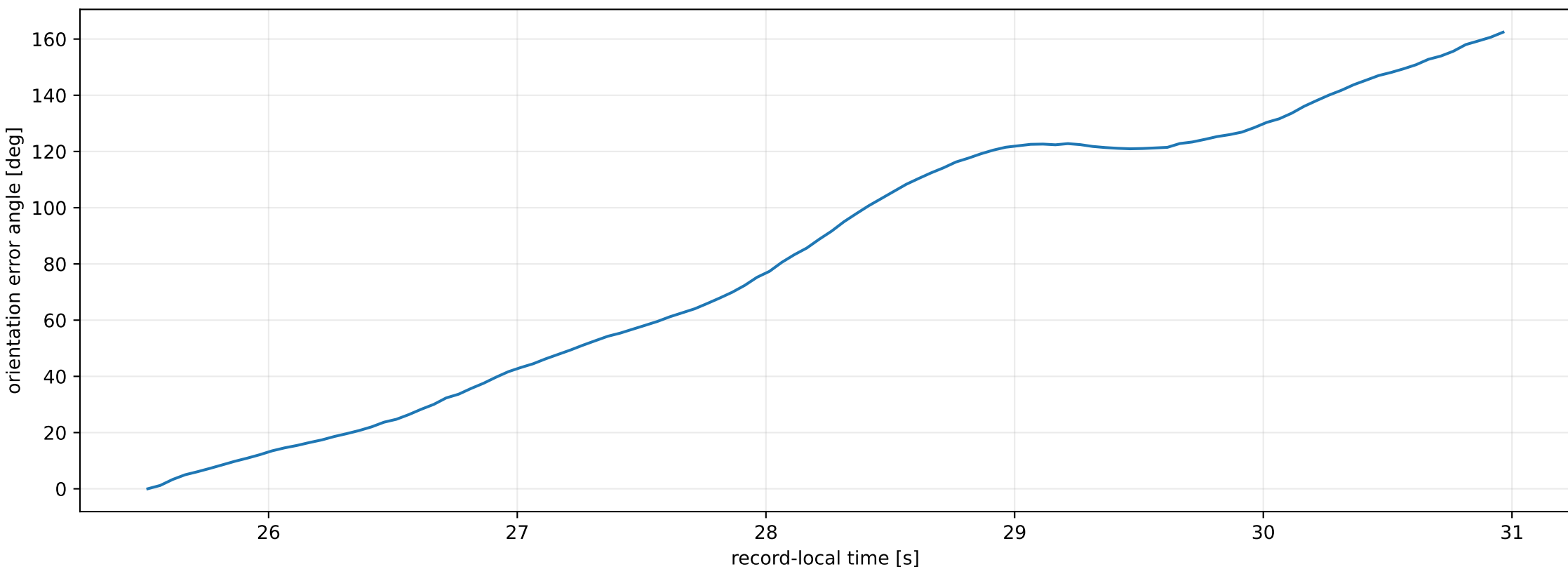
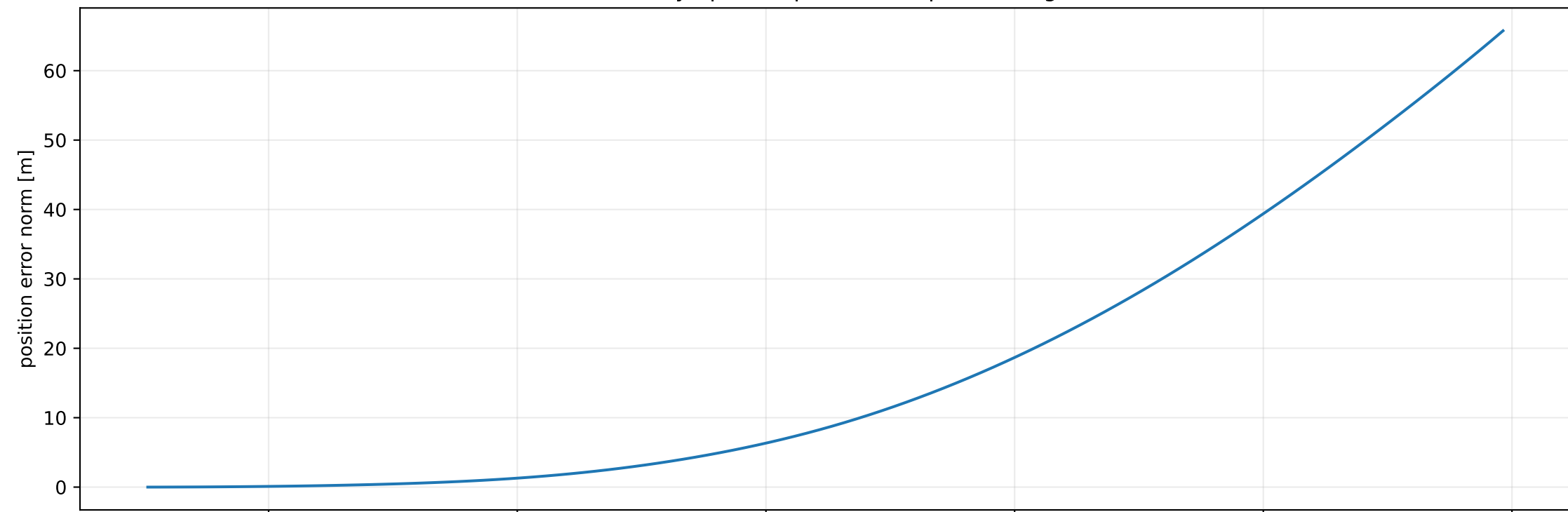
Parameter-only open-loop pose: position



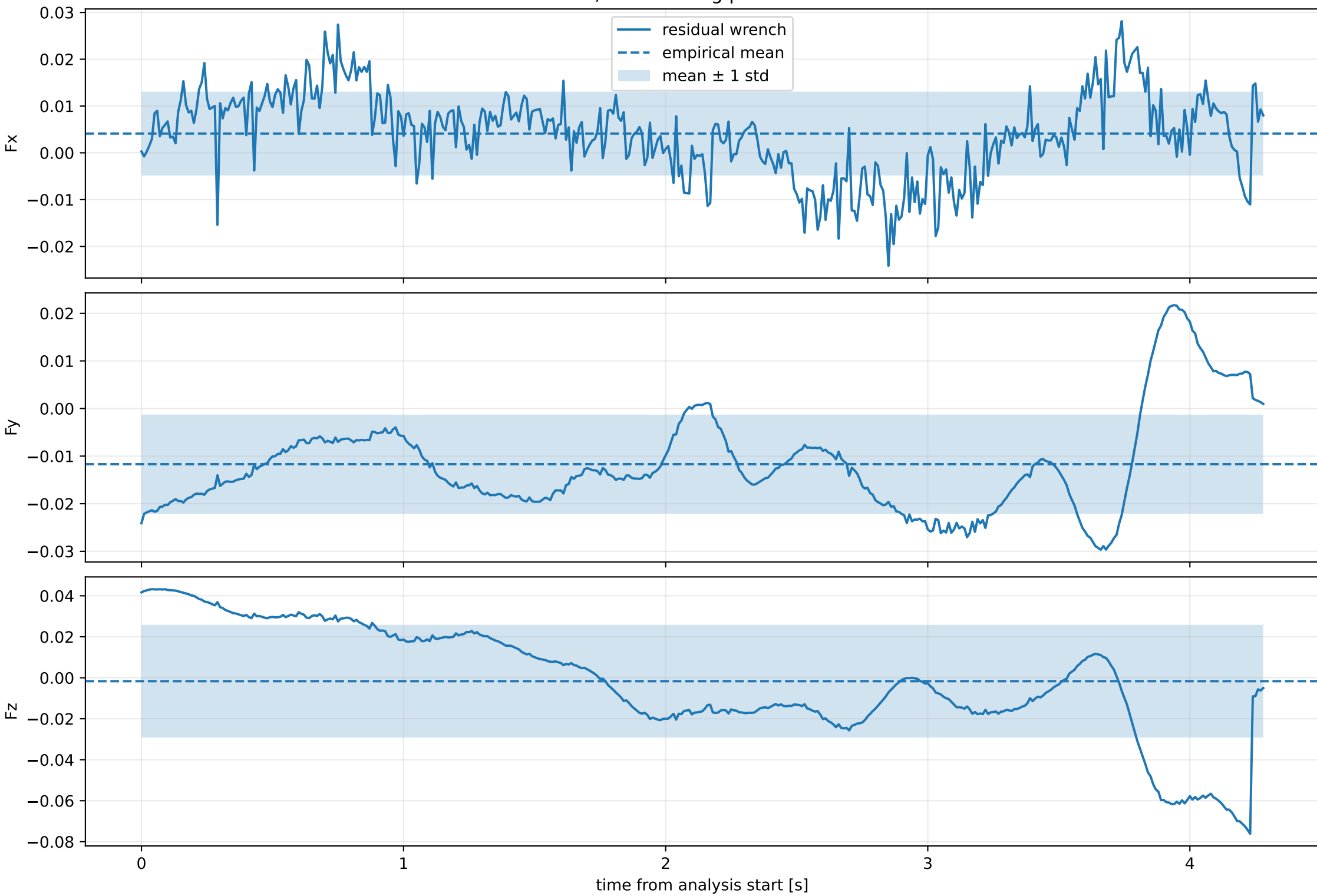
Parameter-only open-loop pose: roll / pitch / yaw



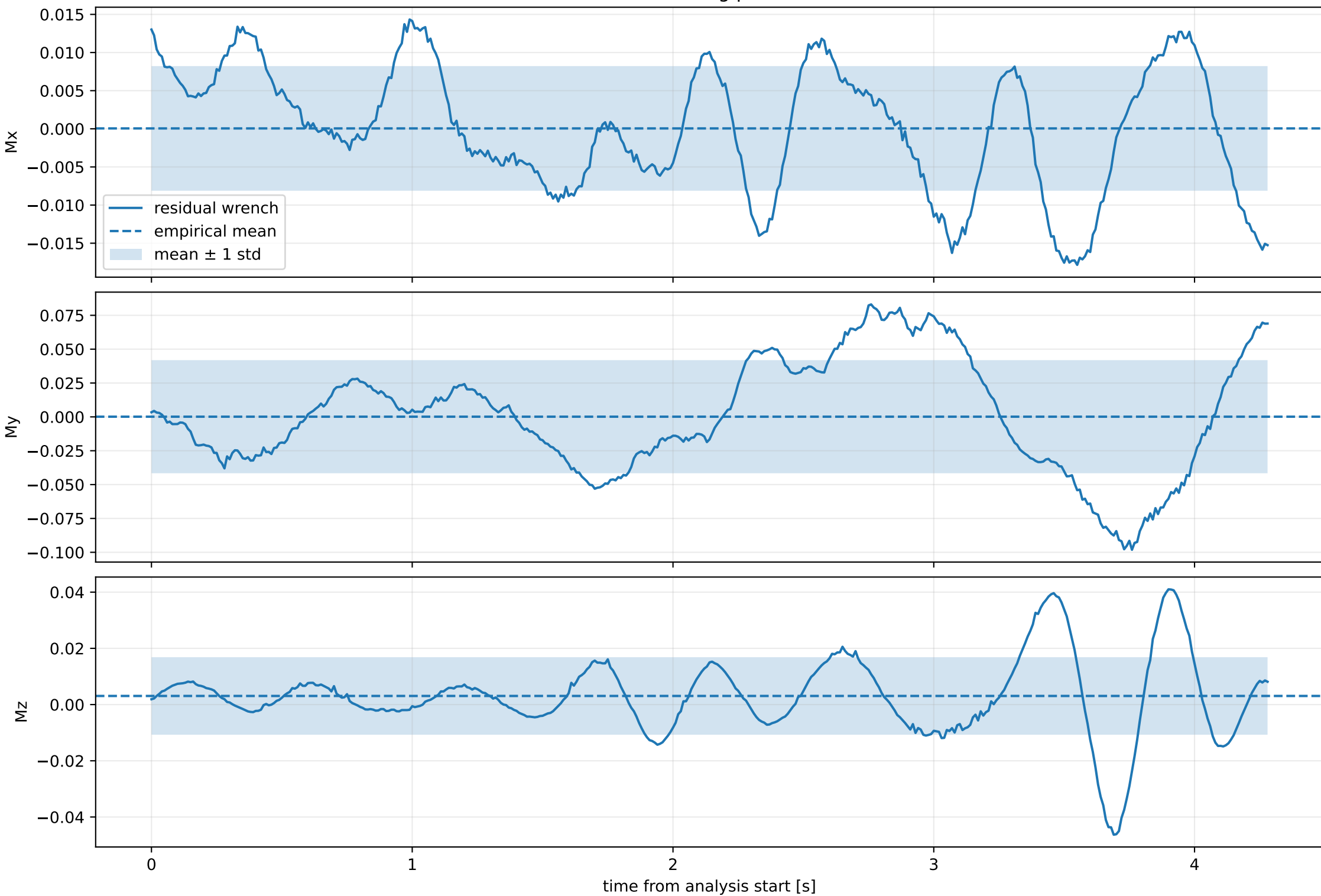
Parameter-only open-loop rollout (separate diagnostic)



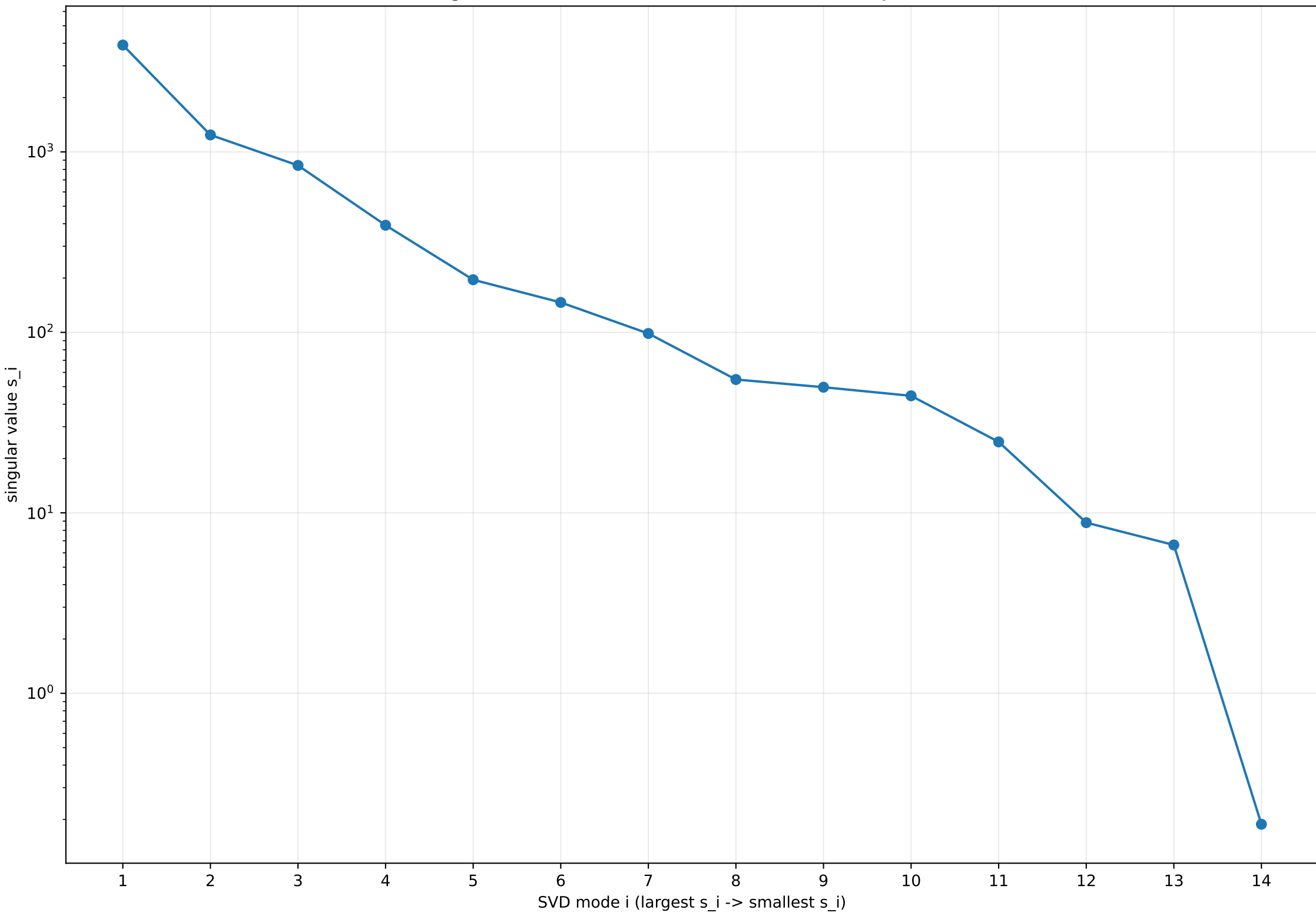
Dimensionless residual force: empirical mean ± 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



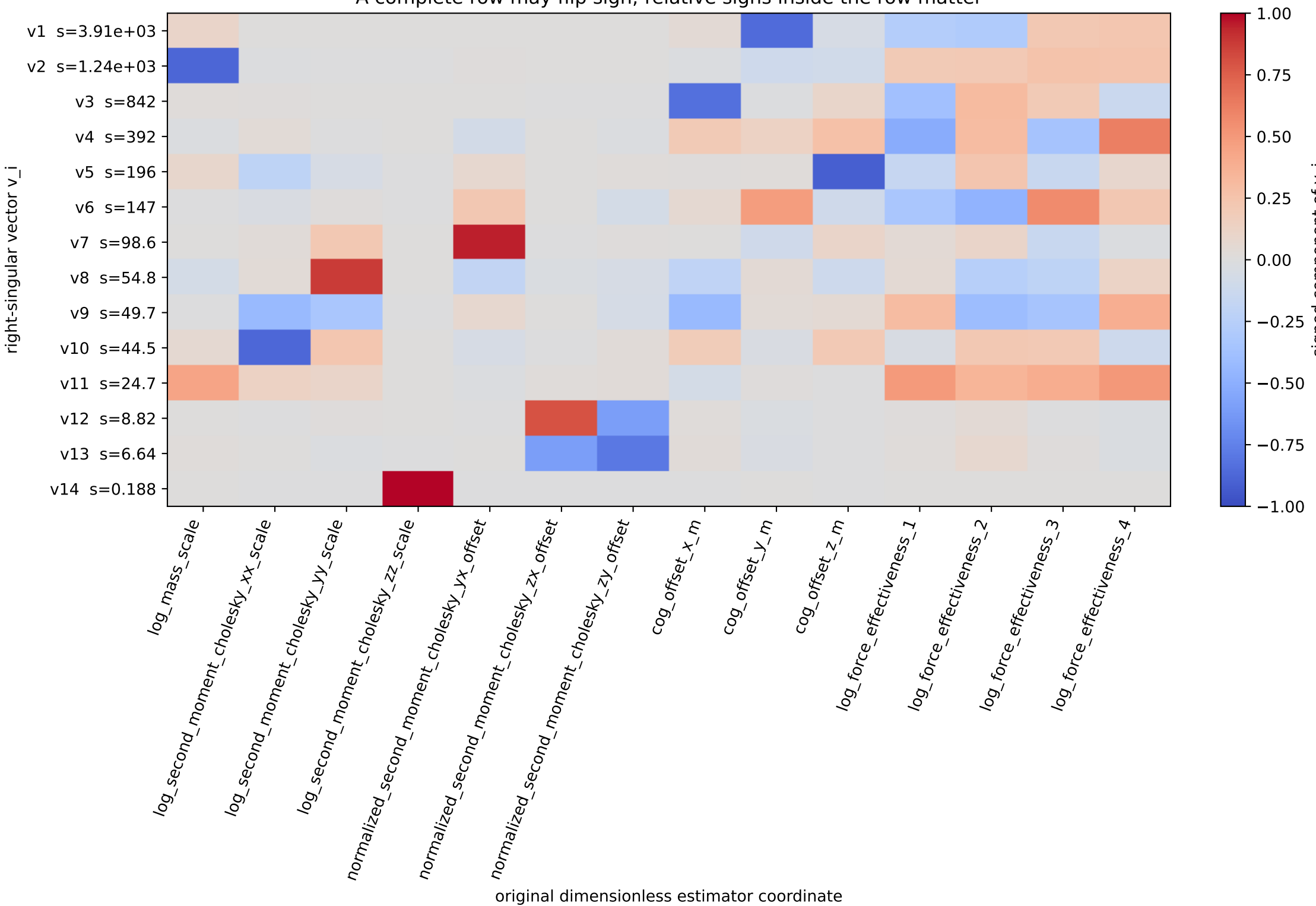
Dimensionless residual torque: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Signed right-singular vectors V^T
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.61776382581

inertia J [kg m²]

[0.066113726 8.9678174e-05 3.3611357e-05]
[8.9678174e-05 0.10803579 1.5843749e-05]
[3.3611357e-05 1.5843749e-05 0.17318655]

principal moments [kg m²]

[0.066113524, 0.10803597, 0.17318657]

CoG offset [m]

[-0.0043561755, 0.0015073299, -0.014263404]

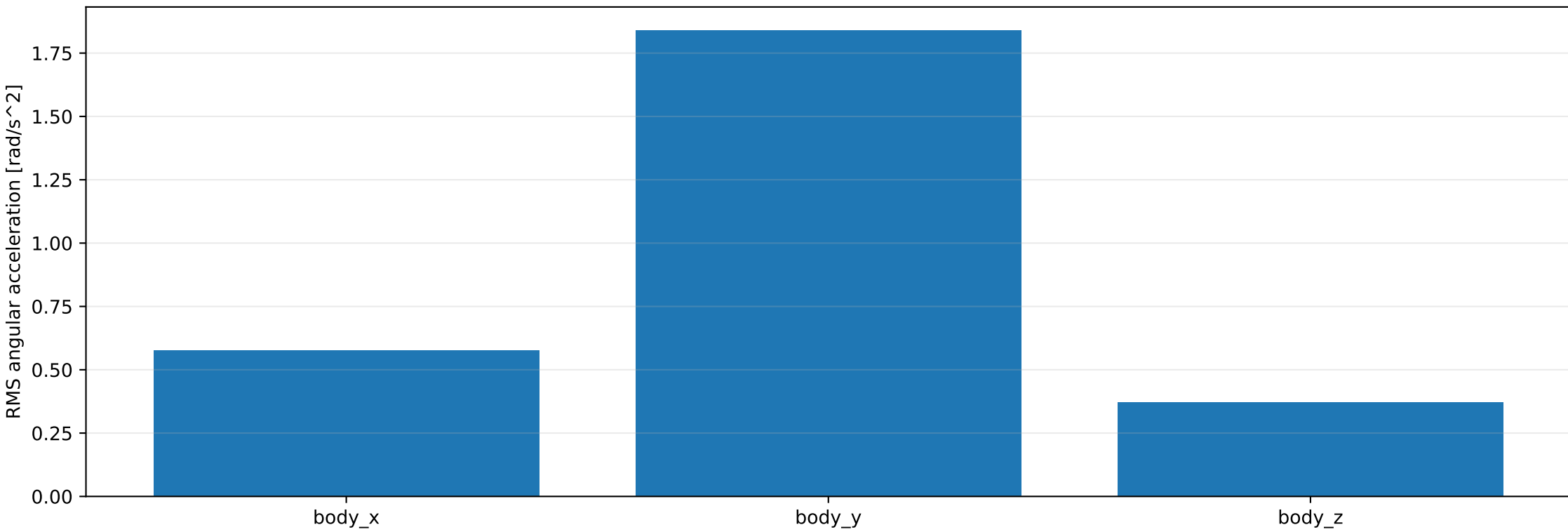
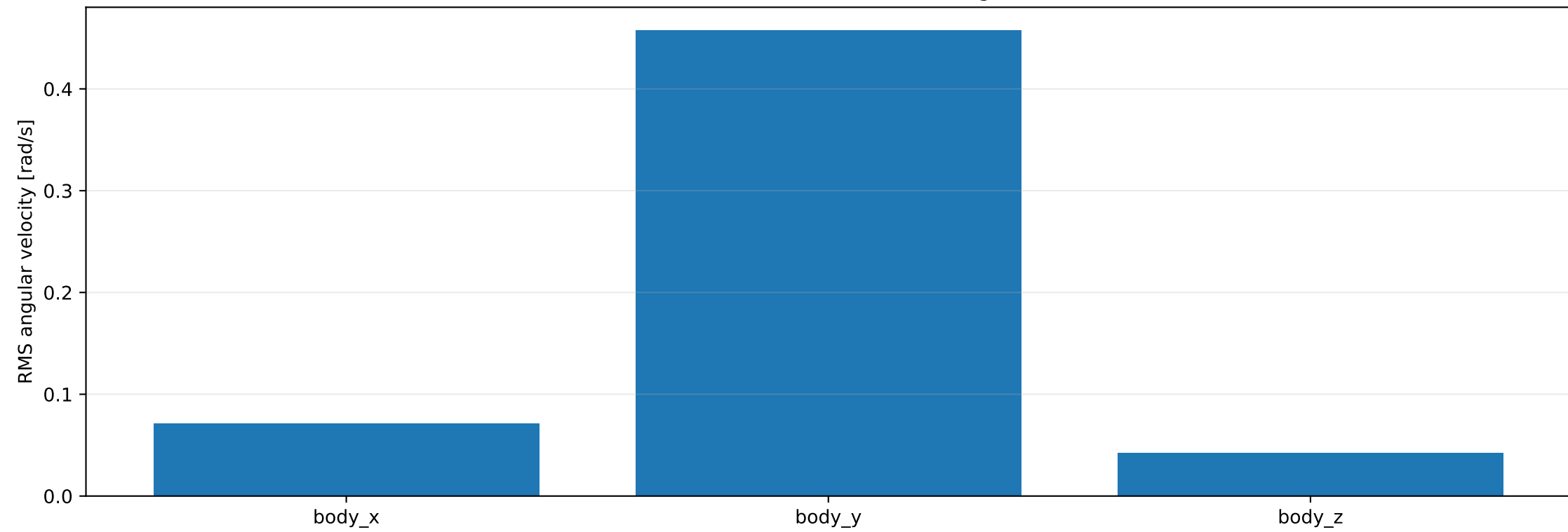
rotor force effectiveness

[0.9495992, 0.94455827, 0.94193517, 0.93102025]

selected delay [s]

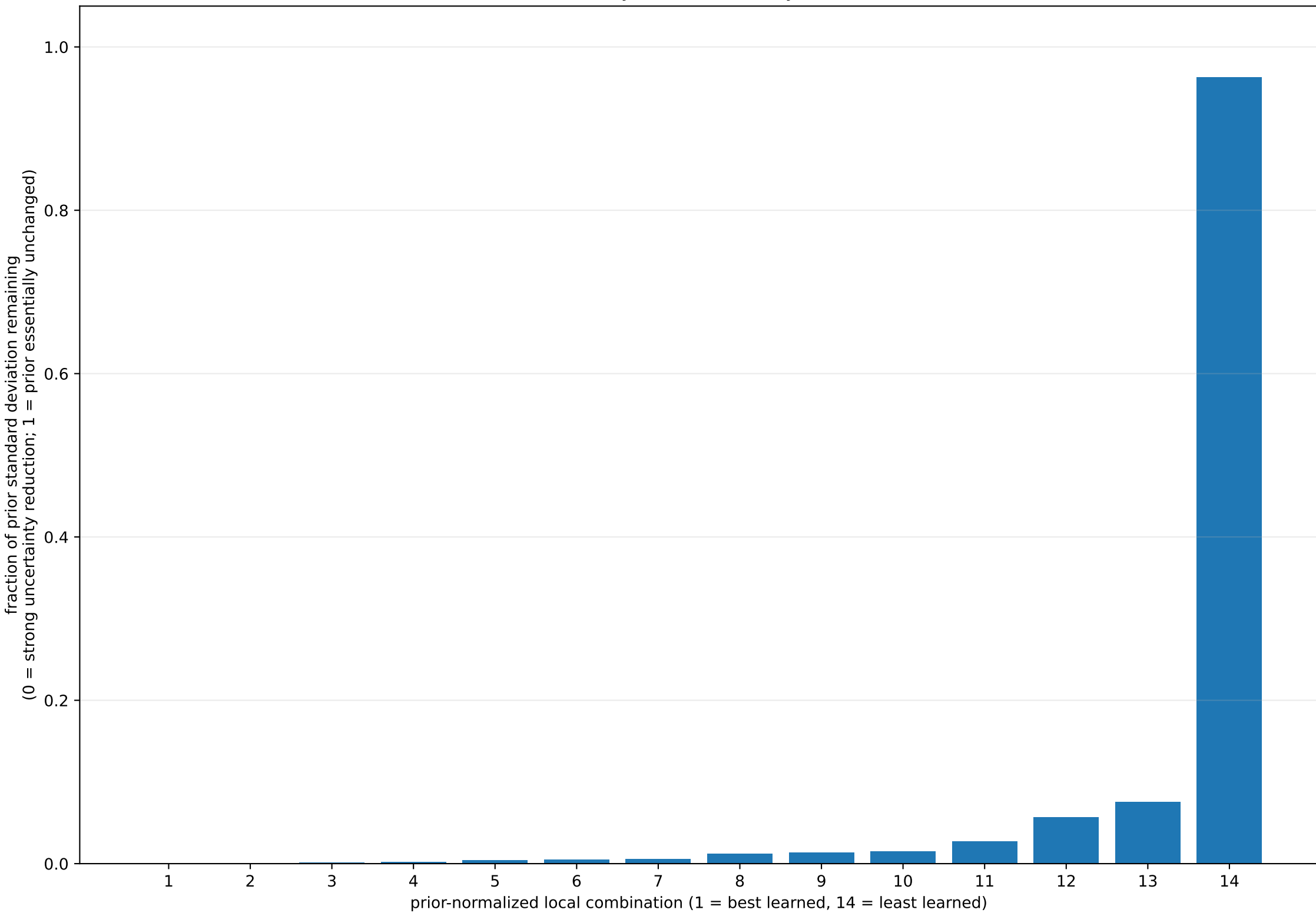
0.046784697252

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

